

Name		Date of Performance	
Roll No.		Date of Submission	
Branch		Examined	

Experiment No:

Title: Measurement of tread parameter using Floating Carriage Micrometer.

Objective:

- 1) To study Floating carriage micrometer
- 2) To measure effective diameter of tread plug gauge
- 3) To measure the major diameter of thread plug gauge
- 4) To study Two Wire method to calculate effective diameter
- 5) To study Best Wire Size

Apparatus:

Instrument	Specifications
Floating Carriage micrometer with standard Specimen method	Steel balls, rolled center type
Tread measuring wire	Diameter 0.895 mm
Master Cylinder	Diameter 10 mm

Screw Thread Terminology

- (1) Screw Thread – A screw thread is a continuous helical groove of specified cross section produced on the external or internal surface of a cylinder or a cone.
- (2) Multiple Start Thread – This is produced by forming two or more helical grooves, equally spaced and similarly formed in an axial section on a cylinder. This gives a quick transverse without sacrificing core strength.
- (3) Crest – Crest is the prominent part of thread i.e, top surface joining the two sides of thread.

- (4) Root – Root is the bottom of the groove between the sides of the two adjacent thread.
- (5) Flank – The straight surface between the crest and the root (which connects the crest with the root) is called the flank.
- (6) Pitch – The distance measured parallel to the axis from a point on a thread to the corresponding point on the next thread is called as pitch of the thread.
- (7) Lead – Lead is the distance through which screw advances axially in one complete revolution. For a single start thread, lead = pitch. For a double start thread, lead = $n \times \text{pitch}$, where n = number of starts.
- (8) Depth of Thread – It is the distance between the crest and the root of the thread measured perpendicular to the axis of the thread. It can also be defined as a distance measured radially between the measure and minor diameters.
- (9) Thread Angle (included angle) – It is the angle included between the flanks or the slopes of the thread measure in the axial plane.
- (10) Flank Angle – The angle made by the flank of a thread with the perpendicular to the thread axis is called flank angle. It is equal to the thread angle.
- (1) Lead Angle – On a straight thread, lead angle is the angle made by the helix of the thread at the pitch line with the plane perpendicular to the axis.
- (2) Helix Angle – On a straight thread, the helix angle is the angle made by the helix of the thread at the pitch line with the axis.
- (3) Major Diameter – It is the diameter of imaginary coaxial cylinder which would touch the crests of an external thread or roots of an internal thread. It is also called as external diameter, core diameter, outside diameter or full diameter of external threads.
- (4) Minor Diameter – It is the diameter of imaginary co-axial cylinder which would touch the roots of the external thread or crests of an internal thread.
- (5) Effective Diameter (pitch diameter) – It is the diameter of an imaginary co-axial cylinder which intersects the flanks of the threads such that the widths of the threads and the width of the spaces between the thread are equal; each being half the pitch.
- (6) Virtual Effective Diameter (functional diameter) – The functional diameter of an external or internal thread is the pitch diameter of an enveloping thread of perfect thread elements having full depth of engagement but clear at crests and roots, and of specified length.

When added to the pitch diameter, the cumulative effects of deviation from specified profile for a specified length of engagement gives the functional diameter.

Measurement of Measure Diameter

- a) Using ordinary micrometer – For the measurement of major diameter of an external thread, a good micrometer is quite suitable. The micrometer is used as a comparator. It is first adjusted in a cylindrical standard of approximate size having approximately the same diameter. This standard is called setting gauge (of size 's')

approximately same as that of the major diameter 'D' of the thread). After taking this reading 'd1' the micrometer set on the major diameter of the thread and the reading is taken 'd2'.

Then the major diameter, $D = S + (d1 - d2)$

Where, S = size of setting gauge

D1 = micrometer reading over setting gauge

D2 = micrometer reading over thread.

As there is a point contact between the crests of the thread and measuring anvils, a light pressure must be applied (do not handle the micrometer ratchet), otherwise error due to deformation of crest may be introduced.

b) **Using Bench Micrometer** – The ordinary micrometer method has deficiencies like variation in measuring pressure, pitch errors in the threads, zero error setting etc. Therefore, for greater accuracy and convenience, a bench micrometer is preferred. This instrument was designed by NPL to remove the deficiencies inherent in the normal hand micrometer.

In order to ensure all that measurements are made at the same pressure, a fiducial indicator is used. This ensures constant pressure for all the measurements. The instrument has a micrometer head with a vernier scale to read the accuracy of 0.002 mm. It is used as a comparator in order to avoid pitch errors of micrometer threads, zero error setting etc. A calibrated setting cylinder having nearly same diameter as major diameter of the thread to be measured is used as a setting standard. The setting cylinder is held between the anvils and reading is taken. In this machine there is no provision for mounting the workpiece between the centres and it is to be held in hand. The cylinder is then replaced by the threaded workpiece and again the micrometer reading is noted.

If, D = diameter of setting cylinder,

R1 = reading of micrometer on setting cylinder,

R2 = reading of micrometer on screw thread

Then major diameter of screw thread, $= D + (R2 - R1)$

Measurement is taken at two or three positions along the thread to determine any taper which may be present, and at least two preferably three angular positions to detect ovality.

Measurement of Effective Diameter

This is the most important of all thread elements as it plays a big role in perfect fitting of the mating threads. The various methods of measuring effective diameter are;

1. Thread micrometer method
2. The one wire, two wire or three wire methods.

Thread micrometer method – This method is simple and rapid. For measuring effective diameter, a thread micrometer is used. The thread micrometer is similar to ordinary micrometer except that it has special contact points to suit the end screw threads form that is to be checked.

The contact points are selected depending upon the type and pitch of the thread to be measured. The male contact point is then fitted into the fix anvil and corresponding female contact point is fitted into the spindle of thread micrometer. For accuracy a setting gauge is used to remove zero error.

The anvils are then made to contact the thread to be checked as shown in fig, and the reading is taken, which gives the effective diameter of the thread. In this method, actually the measurement is between major diameter on one side and minor diameter on the other side which gives the effective diameter.

The reading thus obtained should agree with that obtained by measurements of outside diameter and pitch from the relation; Pitch diameter = $D - 0.6403 p$ (in case of whitworth thread), where $0.6403 p$ = depth of thread, D = outside diameter of workpiece P = pitch of the thread.

The advantage of thread micrometer is that this is the only method which shows the variation of the drunken thread.

Using wires (rods) - This is the most common method of pitch diameter measurement. In this method small rods or wires are on the threads and the measurements are then made over or under the wires with a micrometer or any other more accurate machine. These wires are made of hardened steel to sustain wear and tear in use and are given a high degree of accuracy and finish by tapping to suit different pitches. For each pitch there is a 'best size' of wire. If the best size of wire is used the wire makes contact with the flanks of the thread on the pitch line.

One wire method - In this method only one wire is used. It is placed between two threads at one side and the anvil of the measuring micrometer contacts the crests on the other side. First the micrometer reading d_1 is noted on a standard gauge whose dimension is approximately same as to be obtained by this method. The setting gauge is then replaced by thread under test. Let this reading be d_2 . Then the effective diameter = $D = (d_1 \sim d_2)$

Where D = size of setting gauge

If $d_2 > d_1$ use positive sign above and if

$d_2 < d_1$ use negative sign.

Two Wire Method – In this method the effective diameter of screw thread is measured by placing two wires or rod of identical diameter between the flanks of the thread as shown in fig and measuring the distance over the outside of these wires.

In two wire method wires of suitable size are placed between the standard and micrometer anvils and the first micrometer reading is taken. Let the micrometer reading over standard and wires = R_1 . The standard is then replaced by the screw thread to be measured and second micrometer reading is taken as shown in fig. Let the micrometer reading over screw thread and wires = R_2 .

The diameter of the standard = S

The diameter under the wires = T

The effective diameter of the screw = E

Then $E = T + P$

$$T = S - (R_1 - R_2)$$

And P is a constant calculated for different threads, it depends upon the diameter of wire and pitch of the thread. It, p = pitch of the thread and d = diameter of wire, then $P = 0.9605p - 1.1657d$ (for whitworth threads) and $P = 0.866p - d$ (for metric threads).

Measurement over wires by means of a micrometer is only satisfactory, where high accuracy is not required, therefore, two wire method should be carried out on the diameter measuring machine, because alignment is not possible by two wires and can be provided only by the floating carriage machine.

The machine ensures that the axis of the micrometer is maintained at 90 to the wire of the screw under test. The lower slit moves parallel with the axis of the thread while the top slides moves at 90 to the thread axis.

Three Wire Method – In this method three wires of equal and precise diameter are placed in the thread grooves at opposite sides of the screw and measuring the distance M over the outer surface of the wire with the micrometer. Out of the three wire in the set two wires are placed on one side and third wire on the other side as shown in fig. The wire are either held in hand or secured in the grooves by applying grease in the threads or by sticking the ends of the wires in wax or Vaseline. These wires may also be hung through threads on a stand. Any such method, however, must ensure freedom to the wires to adjust themselves under the micrometer pressure.

This method ensures the alignment of micrometer anvil faces parallel to the thread axis. Therefore, this method of measuring effective diameter is more accurate.

The effective diameter can now be calculated with the following known elements;

1. The reading of the micrometer (this is the distance across the top of wires say M) then,
 $M = E + Q$, where E = effective diameter and Q is the constant depending upon the wire diameter and flank angle.
2. The wire diameter.
3. The thread angle, and $Q = W(1 + \operatorname{cosec} \alpha) - p/2 \cot \alpha$ where W = diameter of wire and p = pitch.

$$\begin{aligned}\text{Thus effective diameter } E &= M - Q \\ &= M - [W(1 + \operatorname{cosec} \alpha) - p/2 \cot \alpha]\end{aligned}$$

Best Wire Size –

Best size of wire is a wire of such diameter that it makes contact with the flanks of the thread on the pitch line. For each pitch there is a best size of wire. Any other size of wire will no doubt contact at the flank but values then obtained for effective diameter will include in it any error due to error in angle.

When the wire makes contact with the flanks of the thread on the pitch line or effective diameter the condition is shown in fig.

The line OP (joining the centers of wire and the contact points) is perpendicular to the flank position of the threads. Let half the included angle of thread be α ,

Then in triangle OAP ,

$$\sin \angle POA = AP/OP$$

$$\begin{aligned}\text{i.e } \sin(90 - x) &= AP/OP \\ \text{or } OP &= AP/\sin(90 - x) \\ &= AP/\cos x = AP \sec x \\ OP &= \text{radius of wire}\end{aligned}$$

Best wire diameter $db = 2AP \sec x$

As AP lies on the pitch line,

$$AP = p/4 \text{ (where } p = \text{pitch of the thread)}$$

$$Db = 2 * p/4 \sec x = p/2 \sec x$$

Observation table

Sr. No.	Parameter	1	2	3	Avg
1	R1				
2	R2				
3	R3				
4	R4				

Where,

R1= Reading of Floating carriage micrometer with standard cylinder without wire.

R2= Reading of Floating carriage micrometer with Screw thread cylinder without wire.

R3= Reading of Floating carriage micrometer with standard cylinder with wire.

R4= Reading of Floating carriage micrometer with Screw thread cylinder with wire.

Calculations:

Conclusion